1. Definitions

Deterministic (Scoped Definition)

“Deterministic” refers to determinism of the governed execution trace, not to probabilistic inference internals. A system is deterministic when identical inputs, identical code-hash, identical measured hardware state, and identical policy state yield byte-identical audit artifacts and replay outputs, regardless of internal stochastic inference.
Governed Execution Envelope
The bounded runtime domain in which all agent actions, model interactions, environmental effects, and external communications are cryptographically constrained, logged, and replayable.
Replay vs. Re-Execution
“Replay” denotes cryptographic reconstruction and verification of a prior execution trace. “Re-execution” denotes deterministic re-driving of agent actions from a recorded state, independent of the original model binaries.

2. System Overview (Reprinted, Clarified)

ROGUE is a local, fully autonomous agentic operating framework that binds:

- deterministic execution governance,
- cryptographically immutable audit trails,
- hardware-rooted trust,
- and legally enforceable human authorization

into a single, court-admissible system.
The framework is designed to satisfy evidentiary standards including FRE 902(13) and compliance regimes such as SOC-2, FedRAMP, GDPR, and the EU AI Act, while operating entirely on-premise.

3. Architecture (Revised with Precision)

Layer 0 : Hardware Root of Trust

- TPM 2.0 measured boot
- PCR-17 extended with:
    - kernel hash
    - agent binary hash
    - policy manifest hash

New Constraint
Any PCR mismatch triggers cryptographic invalidation of execution capability and forces secure NVMe re-imaging from a write-protected medium until the measured state matches the sworn reference state.

Layer 1 Deterministic Runtime Envelope

- Python + Rust execution
- All non-deterministic primitives constrained to:
    - recorded seeds, or
    - logged inference artifacts

NEW VERBIAGE (critical):
No model parameter, adapter, tensor, or inference-affecting state may mutate unless preceded by a signed WAL entry explicitly referencing the mutation hash.

Layer 2 Immutable Flight Recorder (Strengthened)

- Append-only SQLite WAL

- Flush interval ≤ 100 ms
- Each entry contains:
    - mission ID
    - epoch
    - action
    - arguments
    - rationale
    - SHA-256 payload hash
    - Ed25519 signature

NEW
Timestamps MUST derive from a monotonic, hardware-backed or TPM-attested clock source. Any detected clock regression or tampering invalidates the mission.

Layer 3 Ethics & Human Authorization Engine (Closed Loop)

- Deontic logic evaluation
- Policy masks
- Required human authorization gates

NEW (important legal hardening):
Absence of required human authorization results in cryptographic impossibility of execution, not logical refusal. No bypass path exists.

Layer 4 Governance & Equity Permission Lattice

- 1:1 mapping between permission bits and equity or liability classes
- On-chain or ledger-bound authorization records

Layer 5 Multi-Model Sandbox (Clarified)

- Multiple LLMs MAY be used
- Outputs are advisory only
- Execution authority resides exclusively in the deterministic arbiter

NEW
Replay and verification MUST NOT require access to the original model binaries.

Layer 6 Attestation-Gated I/O

- Every outbound request requires:
    - TPM quote
    - time-bound authorization token

NEW
Alternative embodiments may include quorum-based or decentralized authorization oracles to prevent centralization attack vectors.

4. Deterministic Replay & Evidentiary Export (Clarified)

- Reconstructable environment:
    - DOM

    - storage
    - session state
- Merkle-rooted log bundles
- Exportable evidence packs

NEW
Replay artifacts are sufficient for independent verification without trust in the original operator or system owner.

5. Threat Model (Expanded)

Threat	Mitigation
Weight mutation	Pre-signed mutation logs only
Clock tampering	TPM-attested time
Human bypass	Cryptographic veto
Forked binaries	Attestation-gated I/O
Semantic “non-determinism” claims	Scoped definition

6. Conclusion (Rewritten)

ROGUE is not a model wrapper. It is a deterministic governance envelope for autonomy.
By binding hardware trust, immutable logging, human authority, and replayable evidence into a single local system, it establishes a new category:
Autonomous systems that can be audited, replayed, and trusted without belief.

PART II ADDITIONAL CLAIMS (NEW)

Claim 5 Deterministic Governance Scope

A system wherein determinism applies to the governed execution trace such that replay artifacts are byte-identical given identical measured state, irrespective of probabilistic inference internals.

Claim 6 Cryptographic Human Veto

A system wherein absence of required human authorization renders execution cryptographically impossible.

Claim 7 Hardware-Attested Time Integrity

A system wherein all timestamps derive from a monotonic, hardware-backed or TPM-attested time source.

Claim 8 Mutation Pre-Authorization

A system wherein any mutation of inference-affecting state requires a pre-signed, append-only log entry.

Claim 9 Model-Independent Replay

A system wherein replay and verification do not require access to the original model binaries.